

Derivatives of Inverse and Inverse-Trigonometric Functions

AP Calculus AB/BC — Differentiation

Skills practised: Chain rule on arc-functions, the inverse-function derivative, products and quotients, and second derivatives.

Worked pattern. $\frac{d}{dx} \arctan(u) = \frac{u'}{1+u^2}$, so for $y = \arctan(x^3)$ take $u = x^3$, $u' = 3x^2$, and write $y' = \frac{3x^2}{1+x^6}$ — substitute the inner function into the formula, then multiply by its derivative. The same shape works for every arc-function; only the denominator changes.

Leave answers unsimplified only where simplification is genuinely impossible; state exact values (no decimals). Show the inner function u you used.

1. Differentiate: $y = \arctan(3x)$

Answer: _____

2. Differentiate: $y = \arcsin(x^2)$

Answer: _____

3. Let $f(x) = x^3 + 2x + 1$. The function f is increasing, so f^{-1} exists.

(a) Find the value b with $f(b) = 4$. (b) Use $(f^{-1})'(a) = \frac{1}{f'(b)}$ to compute $(f^{-1})'(4)$.

Answer: _____

4. Differentiate: $y = x \arctan(x)$

Answer: _____

5. Differentiate: $y = \arccos(2x)$

(Watch the sign that separates arccos from arcsin.)

Answer: _____

6. Find the second derivative: $y = \arctan(x)$, find $\frac{d^2y}{dx^2}$

Answer: _____

7. Let $g(x) = 2x + \sin(x)$, which is increasing for all x , so g^{-1} exists. Compute $(g^{-1})'(0)$.
(Find the input that g sends to 0 first — no algebra needed for this one.)

Answer: _____

8. Differentiate: $y = e^x \arctan(x)$

Answer: _____

- 9.** Differentiate: $y = \arctan(\sqrt{x})$
(Simplify the denominator completely.)

Answer: _____

- 10. Challenge.** Find the second derivative of $y = \arctan(2x)$, and simplify it to a single fraction.

Answer: _____